

MIT 18.06SC - 2.4

Problem 17.1

Suppose Q is an orthogonal matrix and x is a vector such that $Qx = 0$. Then we have:-

$$\begin{aligned} Q^T Q x &= Q^T 0 \\ I x &= 0 \\ x &= 0 \end{aligned}$$

Since the nullspace is only the zero vector, Q has full column rank i.e. all ~~vectors~~ columns linearly independent.

Problem 17.2

$$\cancel{q_1} = \cancel{a} \quad \cancel{q_1} = \frac{\cancel{a}}{\cancel{\|a\|}}$$

$$A = \frac{a}{\|a\|}$$

$$= \frac{a}{\sqrt{2}} = (1/\sqrt{2}, -1/\sqrt{2}, 0, 0)$$

~~$\|B\|$~~

$$\begin{matrix} B \\ B_{\text{initial}} \end{matrix} = b - \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/2 & -1/2 & 0 & 0 \\ -1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix} - \begin{bmatrix} -1/2 \\ 1/2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} +1/2 \\ 1/2 \\ -1 \\ 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 1/2 \\ 1/2 \\ -1 \\ 0 \end{bmatrix} \div \frac{\sqrt{6}}{2} = \begin{bmatrix} \sqrt{6}/6 \\ \sqrt{6}/6 \\ -\sqrt{6}/3 \\ 0 \end{bmatrix}$$

$$C_{\text{initial}} = \underbrace{C - AA^T C}_{\downarrow} - BB^T C$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 1/2 & -1/2 & 0 & 0 \\ -1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

$$C - AA^T C - BB^T C = \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} - \begin{bmatrix} \sqrt{6}/6 \\ \sqrt{6}/6 \\ -\sqrt{6}/3 \\ 0 \end{bmatrix} \begin{bmatrix} \sqrt{6}/6 & \sqrt{6}/6 & -\sqrt{6}/3 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

Now $\rightarrow$

$$= \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 1/6 & 1/6 & -1/3 & 0 \\ 1/6 & 1/6 & -1/3 & 0 \\ -1/3 & -1/3 & 2/3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \\ -1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1/3 \\ 1/3 \\ 1/3 \\ -1 \end{bmatrix} \div \frac{2\sqrt{3}}{3} = \begin{bmatrix} \sqrt{3}/6 \\ \sqrt{3}/6 \\ \sqrt{3}/6 \\ -\sqrt{3}/2 \end{bmatrix}$$

d is in the left nullspace of both $\{A \ B \ C\}$ and $\{a \ b \ c\}$ as shown below:-

$$\begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 & 0 \\ \sqrt{6}/6 & \sqrt{6}/6 & -\sqrt{6}/3 & 0 \\ \sqrt{3}/6 & \sqrt{3}/6 & \sqrt{3}/6 & -\sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Also, both The dimension of the Column space of both matrices is 3 and since the span of both consists only of

Vectors perpendicular to d . Since the ~~the~~ space of vectors perpendicular to d also has dimension 3, we can conclude that and all vectors in $\{A B C\}$ and $\{a b c\}$ are independent, we can conclude that $\{A B C\}$ and $\{a b c\}$ form a basis for all vectors orthogonal to d .